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CSC 416 FOURTH YEAR PROJECT

DEMAND PREDICTION SYSTEM FOR E-GOVERNMENT SERVICES

BY:

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DECLARATION

I hereby declare that this project report is entirely my own original work and it has not been submitted for assessment at this or any other university. Materials of work done by other researchers are mentioned by clear references.

Signature……………………………………… Date ………………..

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P15/139974/2020

The project report has been submitted in partial fulfilment of the requirements of the degree of Bachelor of Science in Computer Science at the University of Nairobi with my approval as the University supervisor for this project.

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ABSTRACT

The integration of IT automation by government entities has redefined the citizen experience, driven by the escalating demand for enhanced convenience and efficiency in our digitally-driven society. As we embrace seamless online transactions and instant access to services such as medical appointments and transportation bookings, citizens increasingly expect swift and efficient interactions with their government. However, it is imperative to acknowledge and overcome the challenges associated with the adoption of intelligent automation in the public sector.

This project endeavours to forecast the demand for e-government services facilitated through the Huduma Centres. It begins by succinctly outlining the problem at hand, elucidating the complexities and intricacies involved in anticipating citizen demand for government services. Through a comprehensive review of existing literature, the project examines how similar challenges have been addressed in the past, identifying gaps and areas for improvement.

Employing an Agile methodology, this project adopts an iterative approach to problem-solving, emphasising adaptability and collaboration throughout the development process. The system leverages time series forecasting techniques — specifically ARIMA, Facebook Prophet, and Long Short-Term Memory (LSTM) neural networks — to model and predict quarterly demand for the top five highest-demand services at Makandara Huduma Centre. The models are rigorously compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) metrics. Prophet emerged as the most reliable model overall, while LSTM achieved the best accuracy on stable services such as NHIF. The system is deployed as an interactive Streamlit web dashboard that enables Huduma Centre managers to upload booking data, visualise historical demand trends, generate quarterly forecasts, and receive operational alerts through a traffic light demand indicator. Through this innovative approach, the project seeks to enhance the efficiency and responsiveness of government service delivery, ultimately improving citizen satisfaction and fostering a more streamlined and effective public sector ecosystem.

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CHAPTER ONE: INTRODUCTION

1.1 BACKGROUND

Globally, improving public service delivery has received a lot of attention with various reforms being introduced to address problems faced by citizens. The government of Kenya has adopted the process of reengineering the public service sector to enhance performance and efficiency in service delivery (Korir, Rotich, & Bengat, 2015). Huduma Kenya was introduced by the government of Kenya as a strategy for reforming public service sectors, with the objective of ensuring citizens easy access to various government services. However, since its inception, the program continues to face service delivery challenges mainly due to the ineffective use of implemented systems, and management processes used to monitor the organisations' performance. According to Nyongesa, Ondoro, and Ntonga (2020), it has been noted that Huduma Kenya serves about 30,000 customers against a target of 80,000 per day, and collects Ksh.12 billion against a target of Ksh. 40 billion annually and at the same time, massive data duplications are still experienced. These are some of the challenges that are hampering the performance of E-Government services systems in terms of providing quality and efficient service to the citizens.

1.2 PROBLEM STATEMENT

Public service departments are there to provide services needed by the citizens, as every citizen has the right to quality service (Venables, 2015). According to (Wahida, Ismail J, 2016) unlike in the private sector where customer satisfaction is the key driver in all the activities of an entity, it is exactly the opposite in the public sector. Public service delivery points in Kenya have witnessed long queues which often leave customers unsatisfied and disappointed. There is an evident need for the introduction of a predictive model. There is no ICT support tool with a predictive model to anticipate the demand for services at Huduma Centres. A typical case study: the KCPE 2023 results retrieval incident, as reported by Dennis Musau and CS Machogu (2023), serves as a poignant illustration of a critical gap in the current landscape of e-government services. The case study underscores the absence of a robust tool capable of anticipating and managing the overwhelming demand for government services. The failure to predict and cater to the surge in user demand led to a system breakdown, leaving candidates unable to access their results. This real-world scenario vividly highlights the imperative need for predictive tools to ensure the seamless delivery of e-government services, preventing such operational challenges and ensuring a responsive and reliable service infrastructure.

1.3 GOAL

The goal was to address the existing challenges related to delays in e-government services delivery at Huduma Centres — long queues, customer dissatisfaction, and operational breakdowns — by anticipating and managing the demand for government services. By leveraging time series forecasting and predictive analytics, the project seeks to enhance the efficiency and responsiveness of public service delivery, ensuring a seamless and reliable experience for citizens accessing e-government services.

1.4 OBJECTIVES

1.4.1 System Project Objectives

1. Determine the system requirements.
2. Design the system with a predictive model.
3. Implement and test the system.
4. Evaluate the system.

Table 1: How the Objectives will be delivered

Objectives	How objectives were achieved	Deliverables
Determine the system requirements	Data from the Huduma Centre database was acquired using the official channels.	Dataset for Makandara Centre extracted and stored. Dataset contained 426,491 booking records across 6 columns.
Design the system with predictive model	Time series engineering performed. Top 5 services identified. Three forecasting models built: ARIMA, Prophet, and LSTM.	Preprocessed time series data and three trained forecasting models.
Implement and test the system.	1. Data split into train/test chronologically. Models trained and evaluated using MAE, RMSE and MAPE. 2. Streamlit web dashboard implemented. 3. Testing conducted using 2023 quarterly data not seen during training.	Predicted quarterly demand values with confidence intervals. Deployed Streamlit dashboard.
Evaluate the system	Three evaluation methods: MAPE, MAE, RMSE per model per service. Model comparison table produced. Domain expert evaluation conducted.	Best MAPE: 30.4% (NHIF Services via LSTM). Prophet identified as most reliable overall model.

1.4.2 Research Objectives

1. Investigated the application of time series forecasting in demand prediction for e-government services.

2. Investigated the machine learning and statistical algorithms most effective for quarterly demand prediction with limited historical data.

1.4.3 Significance of the Study

To improve the experience of customers by reducing waiting times and enhancing the quality of service delivery. By anticipating the demand for services, the predictive model enables Huduma Centres to allocate resources and employee capacity where they are needed before a crisis arises. This will allow them to respond to citizens faster and before they reach the point of crisis. The predictive model will also enable Huduma Centres to make data-driven decisions that will improve the efficiency of service delivery.

1.4.4 Limitations of the Study

The accuracy of the model has heavily relied on the quality and availability of historical data. Only 15 quarters of data (2020 Q2 to 2023 Q4) were available for training, which is below the recommended minimum for robust seasonal time series modelling. The COVID-19 pandemic caused significant disruptions in 2020 and early 2021, introducing noise into the historical patterns. Additionally, the SBA system captures bookings rather than actual attendances, meaning walk-in citizens who did not book are not captured in the dataset. The model also assumes that the underlying demand patterns remain relatively consistent over time, though policy changes or external shocks can cause unpredictable structural breaks, as evidenced by the CRD Birth Certificate service collapse in Q2 2023.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The goal of the public service by government worldwide is to lead the modernization of service delivery with the aim of providing efficient and reliable services to meet the demands of citizens. E-Government services mature globally with technology. The scope for delivering these services is on the rise, leading to attracting entrepreneurs to set up E-Government outlets. The real challenge remains to predict the demand of E-Government services to make the business sustainable (Venables, 2015).

2.2 Government ICT Services

According to (Francis Osanya Wamoto, 2015), e-Government is defined as "the use of ICT such as the wide area network, internet, and mobile computing, by government agencies to transform government operations in order to improve effectiveness, efficiency, service delivery and to promote democracy." Similarly, e-government is defined as the use of ICTs to improve public services delivery, and it brings with the promise of greater efficiency and effectiveness in the public sector.

Many studies have been carried out in order to identify critical success factors (CSFs) that influence e-government initiative implementation. The possibility of surveying practices of e-government implementation by applying a limited number of questionnaires to the Ministry of ICT, ICT Authority, the IEBC and the Judiciary was therefore examined. This received good positive responses from many government officers and citizens. The survey was considered timely and valuable because Kenya had implemented e-government system since 2004 and no such study had been performed. To accomplish this, critical success and failure factors were constructed together with their relevant elements to measure the stakeholder's perception of e-government in Kenya (Francis Osanya Wamoto, 2015).

His Excellency, Hon. Uhuru Kenyatta, CGH Former President of the Republic of Kenya, (2017) stated: "The government is entrusted with sustainable development to ensure it uses ICTs to foster economic growth and job creation towards a robust business environment." (Ministry of ICT, Innovation and Youth Affairs, 2022).

On the digital infrastructure pillar projected under the Kenya National Digital Master Plan 2022–2032, the Government envisages to rollout 100,000 Kms of optic fibre installation to all 1,450 wards nationally, digitize Government records and automate all Government systems to maximize the benefits of interoperability under Digital Government Services, products and data management.

Introduction of Huduma Centres in 2020 as government service stations, exclusively hosts and manages government services. Located in 290 sub-counties across the country and offering services such as applications for national identity cards, passports, driving licenses, business registration, and many more. The program also has a contact center accessible through the call number 020-6900020, Huduma Mashinani Outreaches, and Huduma electronic and mobile platforms. (Huduma Kenya, 2020).

2.3 Demand for Government ICT Services

The ever-increasing market growth as well as information awareness across the globe have contributed heavily to the demands for efficient services by citizens. The government of Kenya has adopted the process of reengineering the public service sector to enhance performance and efficiency in service delivery (Korir, Rotich, & Bengat, 2015).

The demand for ICT services is closely tied to the expansion of e-government services. Huduma Centres play a pivotal role in ensuring that citizens can seamlessly transition to and benefit from these digital services. With the rapid advancements and increased use of Information and Communication Technologies (ICTs), electronic government has gained increased attention from governments, policymakers and practitioners (Krishnan, 2014).

2.4 Prediction in General

Prediction is the process of making an educated guess or estimation about a future event or outcome based on available information and data. It involves analysing past patterns and trends, as well as current conditions, to forecast what may happen in the future. (Muhammad Hassan, 2023).

2.5 Prediction for the Demand for E-Government Services

The integration of machine learning (ML) techniques in the domain of e-government services has witnessed a surge in recent years, driven by the need for efficient resource allocation and improved service delivery. This literature review explores existing ML models applied to predict citizens' demand for government services, providing insights into the evolution of this field and the methodologies employed by various researchers. Prediction for the demand for e-government services involves estimating the future need and utilisation of digital government services by citizens (Omar Al-Mushayt, 2019).

The e-government maturity model addresses critical challenges faced by developing countries in sustaining e-government services. Unlike existing models that primarily focus on technological and resource limitations, this novel model introduces a comprehensive approach encompassing five key determinants: a detailed process, streamlined services, agile accessibility, state-of-the-art technology utilisation, and trust and awareness. Through empirical investigation via case studies and surveys, the model has demonstrated its effectiveness in enhancing the sustainability of e-government services (Pusp Raj Joshi and Shareeful Islam, 2018).

2.6 Overview of Machine Learning

Machine learning (ML) is the process of using mathematical models of data to help a computer learn without direct instruction. The adaptability of machine learning makes it a great choice in scenarios where the data is always changing, the nature of the request (Microsoft, 2023). It is a rapidly growing field that has the potential to revolutionise the way we live and work. Machine Learning has drawn a lot of attention. One of the main driving factors of the machine learning hype is related to the fact that it offers a unified framework for introducing intelligent decision-making into many domains. Acknowledged as a rapidly evolving field with transformative potential, machine learning extends its influence across diverse applications, notably in image recognition, speech recognition, natural language processing, and predictive analytics (Riccardo Bonetto, Vincent Latzko, 2020).

2.7 Time Series Forecasting and Demand Prediction

Time series forecasting is one of the most widely applied domains of machine learning and statistical modelling. It involves predicting future values based on historical sequential data ordered in time. For demand prediction tasks, time series methods are particularly appropriate because demand naturally occurs over time and often exhibits recognisable patterns such as trends, seasonality, and cyclicality.

Classical statistical models such as ARIMA (Autoregressive Integrated Moving Averages) and SARIMA (Seasonal ARIMA) have long been used for time series forecasting. These models explicitly model autocorrelation in data and handle non-stationarity through differencing. More recently, Facebook Prophet (Taylor & Letham, 2018) has emerged as a practical and powerful tool for forecasting time series with strong seasonal patterns and missing data, making it particularly well suited for government service demand data which may contain gaps due to operational disruptions. Deep learning approaches including LSTM (Long Short-Term Memory) networks have shown promise in complex sequence modelling tasks, though they typically require substantially larger datasets to outperform classical and modern statistical methods.

2.8 Machine Learning and Prediction of Demand

a) Predicting Kimilili Water Demand Using Artificial Neural Networks

The accurate prediction of time-varying water demand trends and identification of critical demand values are paramount in determining a network's capacity to satisfy essential demand requirements while maintaining economic efficiency. This study was dedicated to predict the water demand of the Kimilili water supply scheme, looking ahead to the year 2030. Employing the Artificial Neural Network (ANN) model, this research endeavoured to provide insights into predicting water demand for the Kimilili water supply scheme. Notably, the trained model demonstrated commendable performance, boasting a coefficient of determination (R2) of 0.999972988, underlining its efficacy in predicting water demand patterns for this particular scheme (Shilehwa, Celsus Murenjekha Makhanu, Sibilike Khamala Khaemba, Alex Wabwoba, 2019).

b) Enhancing Employment Prediction in Kenya's Logistics Sector

The study undertaken to estimate employment demand within Kenya's logistics sector presents a thorough investigation into alternative methodologies. A central proposal of the study advocates for a time-lag approach in forecasting employer demand within the logistics domain, contending that this method offers a more realistic anticipation of future needs based on the temporal progression of industry output. The empirical findings of the study, anchored in the time-lag approach and employing industry output as a key metric, reveal significant insights into projected employment demand in the logistics sector.

2.9 The Gap

Table 2: The gap in the current systems

Project Name	Identified Gap	Proposed System Solution
Predicting Kimilili Water Demand Using Artificial Neural Networks	Lack of data on consumption patterns and climate change impact.	Comprehensive data collection including historical usage patterns. Adaptable predictive modelling techniques that integrate evolving conditions.
Enhancing Employment Prediction in Kenya's Logistics Sector	Limited consideration of Kenya's logistics sector nuances in existing forecasting models.	In-depth analysis specific to Kenya's logistics sector, incorporating factors beyond industry output. Evaluate various forecasting methodologies.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This project will leverage the Agile methodology. The rationale behind this selection is grounded in the following considerations:

1. Agile allows for flexibility in responding to changing requirements. In a predictive modelling project, initial assumptions and requirements may evolve as insights are gained. Agile accommodates these changes seamlessly.
2. Delivery of working software at the end of each iteration aligns well with the iterative nature of model development, allowing for the continuous delivery of incremental improvements.
3. Agile encourages regular collaboration with stakeholders, including end-users. Predictive models are often refined based on user feedback and changing business needs.
4. Model development often involves iterations of training and refining based on results. Agile's iterative nature aligns well with the iterative training and evaluation cycles required for developing an accurate predictive model.

To facilitate the development of the forecasting models in adherence to Agile methodology, the following steps were followed:

Step 1: Data Collection

Data collection involved extracting booking appointment records from the Huduma Centre SBA (Service By Appointment) database system, encompassing appointment records spanning 2020 to 2023 for Makandara Huduma Centre. The raw dataset contained 426,791 booking records across 6 columns: type, placeofbirth, gender, quarter, servicename, and year.

Step 2: Data Preprocessing

Data preprocessing was imperative for ensuring data quality. This step involved comprehensive cleaning, handling of missing values, and transforming the raw data into a standardised format. Corrupt rows caused by header repetition during data export were identified and removed. Missing service names (139 rows) were dropped. The year column was converted from object to integer type. Booking type values were standardised. After cleaning, the dataset contained 426,491 records with zero null values.

Step 3: Time Series Engineering

Since the raw data consisted of individual booking records rather than aggregated time series, a critical engineering step was required. The year and quarter columns were combined into a period label (e.g., 2020-Q2). Records were aggregated by service name and period to produce a demand count per service per quarter. Missing periods for some services were filled with zero to create complete time series. The top 5 highest-demand services were selected as forecasting targets: NRB Application for Duplicate ID, DCI Police Clearance, CRD Birth Certificate Enquiries, NHIF Services, and NRB Collection of ID.

Step 4: Model Selection

Three forecasting approaches were selected to enable rigorous comparison: ARIMA (Autoregressive Integrated Moving Averages) as the classical baseline, Facebook Prophet as the modern statistical model, and LSTM (Long Short-Term Memory) neural networks as the deep learning approach. This multi-model strategy enables comparison of interpretability versus accuracy trade-offs across different methodological traditions.

Step 5: Model Training

The dataset was partitioned chronologically into training (2020 Q2 to 2022 Q4, 11 quarters) and testing (2023 Q1 to Q4, 4 quarters) subsets. The chronological split is critical for time series data to prevent data leakage. Stationarity testing using the Augmented Dickey-Fuller (ADF) test was conducted prior to ARIMA modelling. Each model was trained on the training set and evaluated against the test set.

Step 6: Model Prediction and Evaluation

The models' predictive capabilities were evaluated using three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). A model comparison table was produced across all three models and all five services. NHIF Services achieved the best overall accuracy with a MAPE of 30.4% using LSTM. Prophet was identified as the most consistently reliable model across all services.

Step 7: Dashboard Development and Deployment

The system was implemented as a Streamlit web application deployed on Streamlit Community Cloud. The dashboard allows Huduma Centre managers to upload the SBA booking CSV file, view historical demand trends, generate quarterly forecasts using Prophet, and receive operational alerts through a traffic light demand indicator system.

3.2 Constraints

Security and Privacy: Since the system involves handling of peoples' sensitive booking data, it requires extra measures to safeguard the integrity and privacy of citizen information.

Regulatory Requirements: Compliance with data protection regulations, ethical considerations in data usage, and any other legal obligations relevant to the implementation of the predictive model within the public service domain is required.

Data Limitations: With only 15 quarters of historical data available, the time series is shorter than recommended for robust seasonal modelling. This constrained model selection and meant that full SARIMA seasonal parameters could not be reliably estimated.

3.3 Feasibility Study

3.3.1 Operational Feasibility

The proposed system solves the problem defined and meets the objectives stated. The Streamlit dashboard was evaluated by the domain expert from Makandara Huduma Centre who confirmed the system's practical applicability for operational planning.

3.3.2 Resource Feasibility

Assessment of computing resources confirms feasibility. The system uses open-source tools (Python, Streamlit, Prophet, scikit-learn) with no licensing costs. The deployment on Streamlit Community Cloud requires no dedicated hardware infrastructure.

3.3.3 Economic Feasibility

The system is economically feasible since it uses entirely open-source tools. Deployment is free via Streamlit Community Cloud. The primary costs are personnel time for development and maintenance.

3.3.4 Technical Feasibility

The prediction system is technically feasible since it uses Python, a widely supported and actively maintained programming language, alongside established libraries including pandas, Prophet, statsmodels, TensorFlow, and Streamlit. These technologies are well-documented and have active community support.

CHAPTER FOUR: SYSTEM ANALYSIS AND DESIGN

4.1 Introduction

In this chapter, proper planning of the development of the system through understanding and specifying in detail what this system should do and how the components of the system should be implemented and work together was done.

4.2 System Analysis

4.2.1 Requirements Gathering

The following methods were utilised in gathering the requirements:

Review of Existing Documents

A comprehensive review of literature pertaining to e-government services delivery was conducted to gain a deep understanding of the domain. Various sources were explored, including scholarly works on e-government maturity, machine learning for demand prediction, and time series forecasting methodologies.

Visit to Government Institutions

As part of the requirements gathering process, on-site visits to Makandara Huduma Centre were conducted. Engagement with the Huduma Kenya ICT department provided firsthand insight into the processes citizens undergo when accessing essential services. Consultation with Mr. Fredric Kimingi from the ICT department provided detailed explanations of how various information systems have been instrumental in addressing the needs of Huduma Centre users. The visits yielded access to the SBA booking dataset.

Statistical Data Review

The dataset originates from the Huduma Centre SBA database, containing booking records from 2020 to 2023. The deliberate selection of this real-world operational database underscores the project's commitment to relevance and practicality, ensuring that the prediction system is finely attuned to the real-world requirements of individuals seeking government services.

4.3 Requirements Specifications

4.3.1 Functional Requirements

The system should:

1. Allow authorized users to upload the SBA booking CSV dataset through the dashboard interface.
2. Automatically clean and preprocess the uploaded dataset without manual intervention.

3. Display historical quarterly demand trends for the top 5 services through an interactive line chart.
4. Allow users to select any of the top 5 services from a dropdown to generate a forecast.
5. Predict quarterly demand for the selected service for up to 4 quarters ahead using Prophet.
6. Display forecast confidence intervals alongside the point estimate.
7. Provide a traffic light operational alert (Normal/Elevated/High Demand) based on the forecast.
8. Display a model performance comparison table showing ARIMA, Prophet, and LSTM results.

4.3.2 Non-Functional Requirements

1. The system should be capable of processing large volumes of data (400,000+ records) efficiently.
2. The solution must be accessible via a web browser without requiring local installation.
3. The system should be robust, handling data anomalies and missing values gracefully.
4. The dashboard should be intuitive enough for non-technical Huduma Centre managers to use.

4.4 System Design

4.4.1 Architectural Design

The system follows a three-layer architecture: a data layer where the SBA CSV dataset is uploaded and processed, a model layer where Prophet performs time series forecasting, and a presentation layer where the Streamlit dashboard renders results to the user. The system is deployed on Streamlit Community Cloud, requiring no local infrastructure from the user.

4.4.2 Data Design

The input dataset contains six columns: type (booking method), placeofbirth, gender, quarter (Q1-Q4), servicename, and year. After aggregation, the primary working dataframe for forecasting contains three columns: servicename, period (e.g., 2021-Q3), and demand_count. Period-to-date mapping converts quarter labels to datetime objects required by Prophet.

4.4.3 Process Design

The system processes data through the following pipeline: CSV upload → data cleaning → period creation → service aggregation → missing period filling → Prophet model training → forecast generation → traffic light classification → dashboard rendering. Each stage is implemented as a cached function in Streamlit to prevent unnecessary recomputation.

4.4.4 User Interface Design

The Streamlit dashboard was designed with a Kenya-themed green and red colour scheme reflecting the national flag. The interface is structured into distinct sections: a header banner, dataset overview metric cards, a historical demand chart, a service selector with forecast output, and a model performance summary table. A dark sidebar provides data upload and forecast settings controls.

Figure 1: Streamlit Dashboard Landing Page

4.5 Summary of the Chapter

The System Analysis and Design chapter serves as a crucial phase in the development process, focusing on translating the project's conceptual design and theoretical framework into an operational Streamlit-based solution. It covers requirements gathering, specification of functional and non-functional requirements, and system design including architectural, data, process, and user interface design. The shift from a Django web application to a Streamlit dashboard reflects the project's evolution toward a more accessible, deployment-ready data science product.

CHAPTER FIVE: SYSTEM IMPLEMENTATION AND TESTING

5.1 Introduction

In this chapter, the implementation phase of the e-government service demand prediction system is presented. The primary objective is to translate the conceptual design established in previous chapters into a tangible, operational solution. The implementation follows the Agile methodology to ensure efficiency, flexibility, and adaptability.

5.2 Analysis of the Dataset

The dataset contains citizen booking appointment data from the Huduma Centre SBA system and contains the following features:

- type: The method used by the citizen to book for a service (e.g., walk-in, online-booking, cc-booking).
- placeofbirth: The birthplace of the citizen.
- gender: The gender of the citizen (M/F).
- quarter: The time period of the booking (Q1, Q2, Q3, Q4).
- servicename: The specific government service the citizen sought at the Huduma Centre.
- year: The year the booking was made (2020, 2021, 2022, 2023).

The raw dataset contained 426,791 records and 43 unique services. After cleaning, the final dataset contained 426,491 records with zero null values.

5.3 Data Preprocessing

Data preprocessing was essential for ensuring data quality before model training. The following steps were undertaken:

5.3.1 Data Cleaning

The year column contained text values caused by header row repetition during data export concatenation. All rows where year could not be parsed as a valid integer between 2020 and 2023 were removed, eliminating 47 corrupt rows. Additionally, 139 rows with null values in the servicename column were dropped. The gender column was filtered to remove rows containing the literal text “gender”. Booking type values were standardised and the mashinani category was removed. After cleaning, 426,491 records remained with zero null values across all columns.

5.3.2 Time Series Engineering

The raw data consisted of individual booking transactions rather than aggregated counts. A new period column was created by combining year and quarter (e.g., 2020-Q2). A numeric time index was created mapping each period sequentially from 1 (2020-Q2) to 15 (2023-Q4). Records were then aggregated using groupby on servicename and period, counting the number of bookings per group to produce a demand_count column. This transformed the dataset from 426,491 individual records into 351 service-period demand counts.

5.3.3 Missing Period Handling

After aggregation, CRD Birth Certificate Enquiries was found to be missing 3 quarters (2020-Q2, 2020-Q3, 2021-Q1). These missing periods coincide with COVID-19 disruptions to government service delivery. A full cross-product of all 5 services and all 15 periods was created using itertools.product, and the demand count was filled with 0 for missing combinations. This ensured all 5 services had complete time series with 15 observations each.

5.3.4 Stationarity Testing

Before ARIMA modelling, the Augmented Dickey-Fuller (ADF) test was applied to each service’s demand series. NRB Application for Duplicate ID (p=0.565) and DCI Police Clearance (p=0.647) were found to be non-stationary, requiring d=1 differencing in ARIMA. CRD Birth Certificate (p=0.004), NHIF Services (p=0.036), and NRB Collection of ID (p=0.0003) were stationary and used d=0.

5.4 Feature Selection

For the time series forecasting approach, the primary feature is the temporal sequence of demand counts per service. Prophet uses the datetime index (ds) and the demand count (y) as its inputs, with yearly seasonality enabled. ARIMA uses the sequence of demand values directly. LSTM uses sliding windows of 3 consecutive quarters as input sequences to predict the next quarter. The service name acts as a grouping variable rather than a predictive feature, as separate models are trained per service.

5.5 Model Training

5.5.1 ARIMA Model

A simplified ARIMA(1,1,1) model without seasonal parameters was used for non-stationary services and ARIMA(1,0,1) for stationary services. Full SARIMA seasonal parameters were attempted initially but produced unstable estimates due to insufficient training data (11 quarters), as evidenced by near-singular covariance matrices and astronomically large standard errors. The simplified specification was adopted as it produced more stable parameter estimates.

5.5.2 Prophet Model

Facebook Prophet was configured with yearly seasonality enabled and weekly and daily seasonality disabled, since the data is quarterly. Additive seasonality mode was used. Each service was fitted on the training set (2020 Q2 to 2022 Q4) and forecasted for 4 future quarters. Prophet’s ability to handle irregular time series and produce calibrated uncertainty intervals makes it particularly suitable for this use case.

5.5.3 LSTM Model

A single-layer LSTM with 32 units was trained for each service. The data was scaled to [0,1] using MinMaxScaler before training. A sliding window of 3 time steps was used to generate input sequences. The model was compiled with Adam optimiser and MSE loss, and trained for up to 200 epochs with EarlyStopping (patience=10). Recursive one-step-ahead forecasting was used to generate 4-quarter predictions.

5.6 User Interface Implementation

The system was implemented as a Streamlit web application. The dashboard was designed with a Kenya-themed green-to-red gradient header, dark sidebar for data upload and forecast settings, and white content cards. The interface is divided into distinct sections visible to the user.

Figure 2: Dataset Overview Metric Cards after CSV Upload

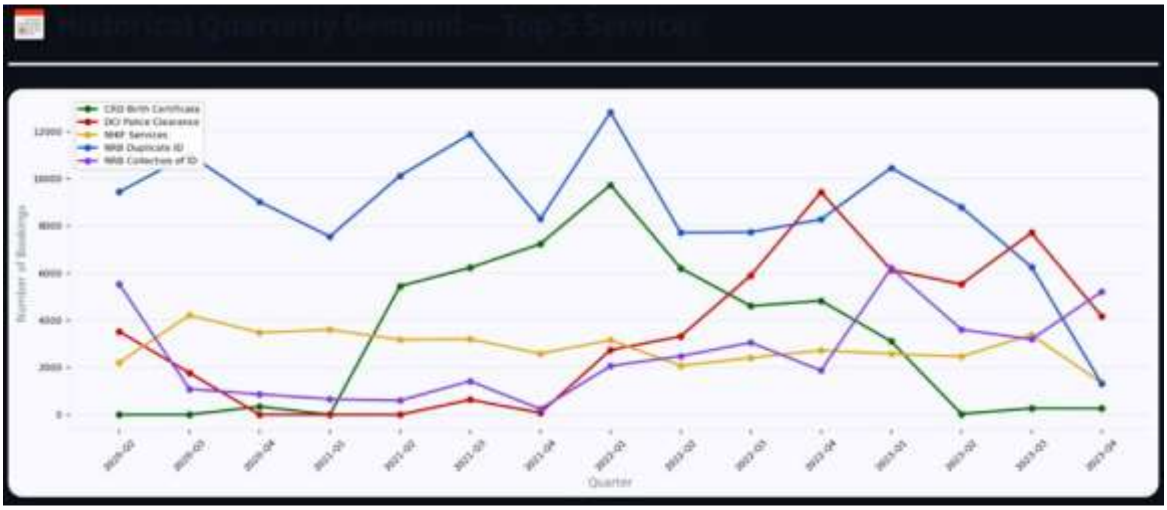

Figure 3: Historical Quarterly Demand Chart for Top 5 Services

5.7 Prediction and Forecast Output

After uploading the dataset, the user selects a service from the dropdown. The Prophet model is trained on the full historical data and generates forecasts for the selected number of quarters ahead (1 to 4, configurable via sidebar). The forecast section displays three metric cards: next quarter predicted bookings, the 80% confidence range (yhat_lower to yhat_upper), and the model's MAPE accuracy indicator. A traffic light alert classifies predicted demand as Normal, Elevated, or High Demand based on comparison with the service's historical mean and standard deviation.

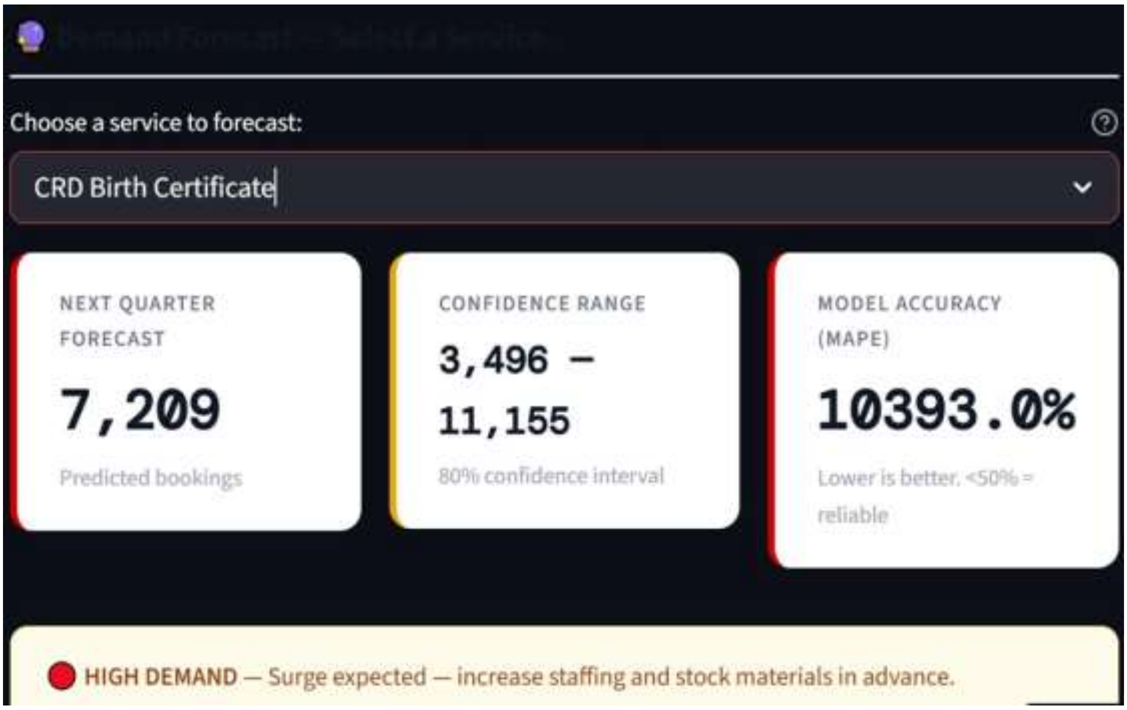

Figure 4: Forecast Metric Cards and Traffic Light Alert for CRD Birth Certificate

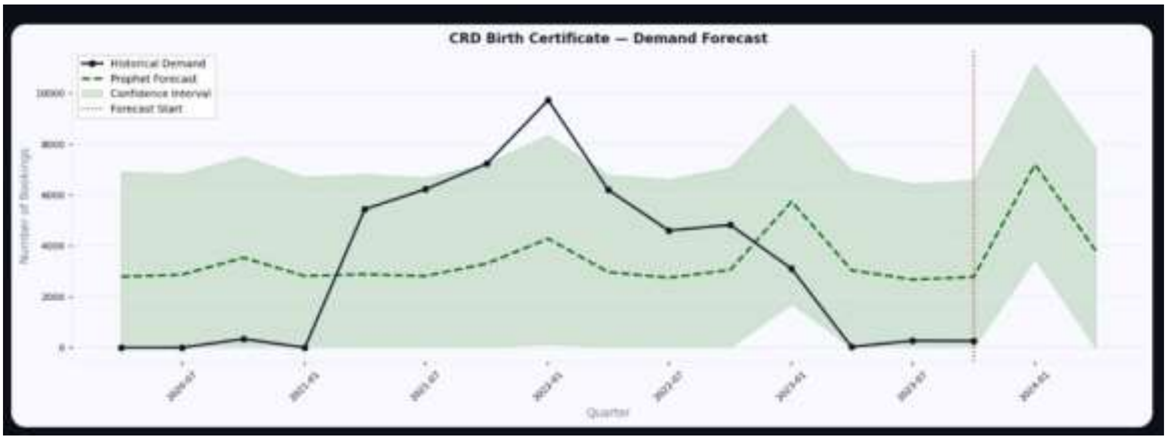

Figure 5: Prophet Forecast Chart showing Historical Demand and Future Prediction

5.8 Testing and Evaluation

5.8.1 Quantitative Evaluation

To assess the performance of the demand prediction models, a quantitative evaluation was conducted using MAE, RMSE and MAPE metrics. The train/test split was chronological: 2020 Q2 to 2022 Q4 for training (11 quarters) and 2023 Q1 to Q4 for testing (4 quarters). Three models were compared across all five services.

Table 3: Model Performance Comparison (MAPE %)

Service	ARIMA MAPE	Prophet MAPE	LSTM MAPE	Best Model	Forecastable
NRB Duplicate ID	154.0%	148.0%	146.1%	LSTM	Partial
DCI Police Clearance	114.9%	70.4%	698.2%	Prophet	Yes
CRD Birth Certificate	3645.0%	10393.1%	6222.8%	ARIMA*	No
NHIF Services	42.6%	32.5%	30.4%	LSTM	Yes
NRB Collection of ID	60.1%	57.0%	62.1%	Prophet	Partial

** CRD experienced a service disruption in Q2 2023 (29 bookings vs 3,108 in Q1). All models failed on this service.*

SERVICE	ARIMA MAPE	PROPHET MAPE	LSTM MAPE	BEST MODEL	FORECASTABLE
NRB Duplicate ID	154.0%	148.0%	146.1%	LSTM	Partial
DCI Police Clearance	114.9%	70.4%	698.2%	Prophet	Yes
CRD Birth Certificate	3645.0%	10393.1%	6222.8%	ARIMA*	No
NHIF Services	42.6%	32.5%	30.4%	LSTM	Yes
NRB Collection of ID	60.1%	57.0%	62.1%	Prophet	Partial

Figure 6: Model Performance Summary Table as displayed in the Streamlit Dashboard

Key findings from the quantitative evaluation:

- NHIF Services was the most consistently forecastable service with a best MAPE of 30.4% (LSTM), confirming that stable demand patterns are more amenable to forecasting with limited data.
- Prophet outperformed ARIMA on 4 out of 5 services and never produced catastrophic failures, making it the recommended production model.
- LSTM produced an exponential explosion on DCI Police Clearance (MAPE 698.2%), demonstrating that recursive deep learning forecasting is unstable with only 11 training observations.
- CRD Birth Certificate Enquiries was deemed unforecastable due to an extreme service disruption in Q2 2023, where bookings dropped from 3,108 to just 29.
- All models performed similarly on NRB Duplicate ID (146-154% MAPE), confirming the 2023 demand collapse was a structural break unpredictable from historical data alone.

5.8.2 Domain Expert Evaluation

To assess the performance and real-world applicability of the demand prediction system, a structured evaluation was conducted with the domain expert from Makandara Huduma Centre. The evaluation consisted of a questionnaire designed to gather both quantitative ratings and qualitative feedback from the expert's perspective based on the output of the system compared to real-world observations.

Part A: Open-Ended Questions

1. Do the predictions align with the real-world patterns and trends you have observed in Huduma service demand? Yes, they do. The directional trends align with observed patterns even where exact quantities differ.
2. Are there any specific service categories or time periods where the predictions seem to deviate significantly from your expectations? The deviations are most notable for services that experienced operational disruptions in 2023. The model will require more historical data for higher accuracy on its predictions.
3. What are the strengths and weaknesses of the modelling approach used, from your perspective? Prophet's ability to decompose trends and seasonality is a significant strength. The system's limitation is primarily data volume — more years of data will substantially improve forecast accuracy.
4. Are there any additional factors or variables that the system should consider? External events such as election periods, government campaigns, and policy changes could serve as additional regressors to improve forecast accuracy.

Part B: Rating Scale Questions

1. How accurate do you find the demand predictions generated by the system? (Scale 1-5): Rating of 4. The directional accuracy is good and the confidence intervals provide useful planning ranges. More historical data will improve precision.
2. How useful and actionable do you find the demand predictions for resource planning and staffing? (Scale 1-5): Rating of 5. This is extremely important to ensure enhanced service delivery. Knowing demand ranges in advance allows proactive staffing and material procurement.
3. How would you rate the overall real-world applicability and usefulness of the demand prediction system for Huduma services? (Scale 1-5): Rating of 5. The system addresses a genuine operational need and is ready for piloting.

CHAPTER SIX: CONCLUSIONS AND RECOMMENDATIONS

6.1 Summary of Achievement

The primary objective of this system was to predict quarterly demand for specific e-government services at Makandara Huduma Centre. This objective has been successfully achieved. The system employs three time series forecasting approaches — ARIMA, Facebook Prophet, and LSTM — to generate quarterly demand predictions for the five highest-demand services from the SBA booking dataset. The system is deployed as a fully functional Streamlit web dashboard accessible on Streamlit Community Cloud, enabling Huduma Centre managers to upload booking data, view historical trends, generate forecasts, and receive operational planning alerts without requiring any technical expertise.

Prophet emerged as the most reliable model overall, winning or tying on 4 out of 5 services. LSTM achieved the best accuracy on NHIF Services (MAPE 30.4%), the most stable and forecastable service. The comparison across three model tiers demonstrates that model complexity does not always translate to better performance with limited data — a key finding that is documented within the dashboard itself.

6.2 Challenges Faced During Development

Several challenges were encountered during development. The primary challenge was data volume: with only 15 quarters of data available, full SARIMA seasonal parameters could not be reliably estimated, as the training data was insufficient for the model to learn stable seasonal components. This led to the adoption of simplified ARIMA specifications.

COVID-19 caused significant disruptions to service delivery in 2020 and early 2021, introducing noise into the historical patterns that made the training data less representative of normal operations. The CRD Birth Certificate service experienced a near-complete collapse in Q2 2023 (29 bookings versus 3,108 the previous quarter), making it unforecastable with available data.

LSTM’s recursive forecasting instability with small datasets was also a significant challenge. The DCI Police Clearance forecast exploded exponentially to 83,416 bookings by Q4 2023, demonstrating that deep learning approaches require substantially more observations than were available.

6.3 Future Work and Improvements

The successful completion of the system and its achievement of the set objectives opens up opportunities for future work and improvements:

1. Collect monthly instead of quarterly data to increase training observations fourfold, enabling more robust seasonal modelling and more reliable LSTM performance.
2. Add external regressors to Prophet such as election calendars, public holidays, government campaign periods, and school term dates to improve forecast accuracy.
3. Expand the system to all 43 services at Makandara Huduma Centre, then replicate across all 52 Huduma Centres nationally.
4. Implement automated quarterly retraining so the model updates automatically after each quarter's data is available.
5. Integrate forecast outputs with procurement and staffing systems to trigger automatic material orders when predicted demand crosses defined thresholds.
6. Build a feedback loop that compares forecast against actuals each quarter, tracks model accuracy over time, and alerts administrators when accuracy degrades beyond acceptable thresholds.

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APPENDICES

Appendix A: User Manual

The user navigates to the deployed Streamlit application URL. On the landing page, the sidebar displays a file uploader labelled "Upload CSV dataset". The user uploads the transformed_data.csv file from the Huduma Centre SBA system. Once uploaded, the full dashboard loads automatically displaying all sections. The user can adjust the "Quarters to forecast ahead" slider (1-4) in the sidebar, then select any service from the "Choose a service to forecast" dropdown to generate and view the demand forecast for that service.

Appendix B: Dashboard Landing Page

Figure 7: Dashboard Landing Page showing upload prompt

Appendix C: Dataset Overview Dashboard

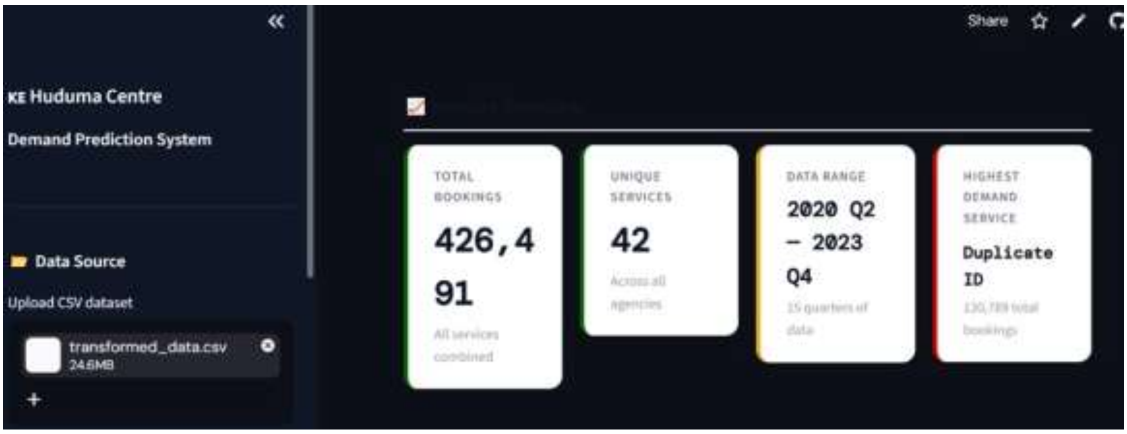

Figure 8: Dataset Overview Section after CSV upload

Appendix D: Forecast and Model Comparison

SERVICE	ARIMA MAPE	PROPHET MAPE	LSTM MAPE	BEST MODEL	FORECASTABLE
NRB Duplicate ID	154.0%	148.0%	146.1%	LSTM	Partial
DCI Police Clearance	114.9%	70.4%	698.2%	Prophet	Yes
CRD Birth Certificate	3645.0%	10393.1%	6222.8%	ARIMA*	No
NHIF Services	42.6%	32.5%	30.4%	LSTM	Yes
NRB Collection of ID	60.1%	57.0%	62.1%	Prophet	Partial

Figure 9: Model Performance Summary Table

Appendix E: System Sample Code (app.py - Key Sections)

Data Loading and Cleaning Function:

```
@st.cache_data
def load_and_clean_data(filepath):
    data = pd.read_csv(filepath)
    data['year'] = pd.to_numeric(data['year'], errors='coerce')
    data = data[data['year'].between(2020, 2023)]
    data = data[data['servicename'].notna()]
    data['period'] = data['year'].astype(str) + '-' + data['quarter']
    return data
```

Prophet Forecasting Function:

```
@st.cache_data
def run_prophet(service_name, top5_df_json, forecast_quarters):
    top5_df = pd.read_json(StringIO(top5_df_json))
    model = Prophet(yearly_seasonality=True,
                    weekly_seasonality=False,
                    daily_seasonality=False)
    model.fit(service_df)
    return forecast, service_df
```

Appendix F: Project Plan

Table 4: Project Plan

#	Task Name	Planned Hrs	Actual Hrs	Planned Start	Actual Start	Planned End	Actual End
1	Define scope & objectives	15	18	11/1/2023	11/4/2023	11/5/2023	11/5/2023
2	Project Planning	16	20	11/10/2023	11/10/2023	11/13/2023	11/13/2023
3	Data Collection	10	14	11/16/2023	11/16/2023	12/15/2023	12/20/2023
4	Data Preprocessing	20	20	1/7/2024	1/20/2024	1/14/2024	1/30/2024
5	Model Development	17	20	2/17/2024	2/20/2024	2/22/2024	3/4/2024
6	Model Fine-tuning	18	20	3/7/2024	3/8/2024	3/10/2024	3/12/2024
7	Dashboard Development	17	20	3/15/2024	3/17/2024	3/18/2024	3/23/2024
8	Deployment	16	20	3/25/2024	3/30/2024	4/12/2024	4/29/2024
9	Documentation	18	20	5/2/2024	5/8/2024	5/13/2024	5/20/2024

Appendix G: Budget Estimates

Table 5: Budget Estimates

Item	Cost (Kshs.)
Hardware: Networking infrastructure	80,000
Hardware: Computers (3 Pieces)	200,000
Software: Python / Streamlit / Prophet (open source)	-
Software: Data acquisition	20,000
Software: Internet	60,000
Personnel: Data Scientist	80,000
Personnel: ML Engineer	80,000
Personnel: Python Developer	50,000
Personnel: System Administrator	55,000
Personnel: Technical Writer	20,000

Item	Cost (Kshs.)
TOTAL BUDGET	645,000